NTK for Low-Rank and Random Features Neural Network

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Abstract

This paper develops a comprehensive Neural Tangent Kernel (NTK) theory for low-rank random feature neural networks, establishing these architectures as a principled route to the kernel regime with enhanced theoretical guarantees and computational advantages. Our main result demonstrates that low-rank random features preserve the same reproducing kernel Hilbert space expressivity as fully-connected networks while achieving kernel behavior at dramatically smaller parameter counts. This linear-width scaling fundamentally lowers the entry barrier to the NTK regime compared to dense architectures, enabling accurate kernel predictions with orders-of-magnitude fewer parameters.

Beyond expressivity equivalence, we establish exceptional concentration properties: kernel fluctuations concentrate exponentially faster in rank than classical NTK theory predicts, making the deterministic kernel limit accessible at moderate model sizes. The spectral structure at initialization exhibits a clean bulk–outlier decomposition consistent with random matrix theory, where a single dominant eigenvalue governs early training dynamics while the bulk explains conditioning properties. Our analysis leverages Fisher–Kibble decoupling to avoid complex correlation diagram calculations, providing a streamlined theoretical framework that remains rigorous yet tractable.

We develop a systematic finite-width corrections program that organizes deviations from the infinite-width limit through a layered expansion, enabling precise characterization of how depth and rank interact in kernel approximations. The architectural simplicity of training only output weights while freezing feature weights eliminates weight-particle coupling between layers, simplifying mean-field analysis while preserving universal approximation throughout training. These results provide a complete theoretical foundation for understanding low-rank random feature networks in the lazy training regime, with concrete practical guidance emerging naturally from the analysis.

Keywords Neural Tangent Kernel · Random Features · Low Rank · Scaling Laws · Infinite Width

Contents

*This work, including all theoretical derivations, calculations, and manuscript writing, was completed by J.H-A. during an internship at UMD.

1 Introduction

Multi-Component and Multi-Layer Neural Networks (MMNNs), also known as low-rank random feature models (RF-LR), represent a promising architecture that combines the expressive power of deep networks with remarkable computational efficiency [1]. These architectures introduce low-rank bottlenecks at each layer, alternating between wide hidden layers of width N and narrow bottleneck layers of rank $r \ll N$, where only the output weights $A^{(\ell)}$ are trained while the input weights $w^{(\ell)}$ remain frozen as random Gaussian draws. Recent empirical work by Zhang *et al.* [1] has demonstrated that MMNNs achieve **global strong convergence** across a wide range of tasks, exhibiting striking phenomena including super convergence and stepwise loss dynamics with long plateaus followed by sharp drops.

Despite these empirical successes, a fundamental theoretical understanding of **how the loss landscape behaves** in MMNNs remains incomplete. The remarkable convergence properties observed in practice—where training displays emergent staircase-like behavior and achieves global minima even when standard neural network training would fail—suggest that the landscape structure of MMNNs differs fundamentally from that of fully-trainable networks. Understanding these landscape properties requires developing a theoretical framework that can explain why MMNNs exhibit such favorable optimization behavior, what drives the stepwise dynamics, and how the low-rank structure shapes the loss surface geometry.

The Neural Tangent Kernel (NTK) framework [2, 3, 4, 5, 6] provides a natural lens for analyzing the lazy training regime, characterizing infinitely wide networks as kernel methods with a fixed data-dependent kernel. This perspective has been extended to finite width and depth, revealing how kernel structure governs training dynamics and generalization. However, applying NTK theory to MMNNs raises new challenges: the frozen random features create a fundamentally different correlation structure, the low-rank bottlenecks modify the parameterization, and the empirical success in achieving global convergence suggests that the landscape properties differ from classical NTK predictions.

Low rank architecture

The considered architecture is a stacking of high then low width layers:

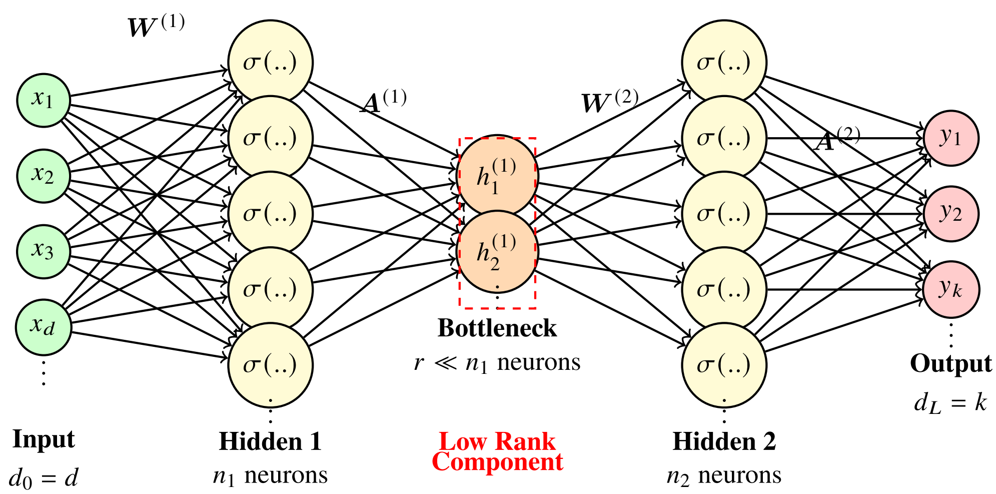

Figure 1: Architecture of a Multi-component Multi-layer Neural Network (RF-LR). The red dashed connections indicate the low-rank component that reduces dimensionality from n_1 to $r \ll n_1$ neurons. We **only train the output weights** $\boldsymbol{A}^{(1)}$ and $\boldsymbol{A}^{(2)}$.

Our work builds a theoretical framework to understand how the loss landscape behaves in MMNNs/LR-RF, focusing specifically on the NTK regime where analytical tractability is greatest and the advantages of the low-rank random feature structure are most clearly visible. By characterizing the NTK structure, spectral properties, and RKHS geometry, we provide a foundation for understanding the favorable optimization dynamics observed empirically. This theoretical foundation is essential for explaining why MMNNs/LR-RF achieve global convergence, understanding the mechanisms behind stepwise loss dynamics, and predicting when and how the low-rank structure confers advantages over fully-trainable architectures.

This work addresses the following fundamental questions:

- *How can we develop a theoretically tractable NTK analysis for MMNNs/LR-RF that explains their landscape properties and guides principled finite-width corrections ?*

Our approach develops an NTK theory specialized to low-rank random feature architectures, leveraging the decoupling properties of frozen random features to simplify the analysis. By exploiting Fisher–Kibble distributions [7, 8] and the independence structure inherent in frozen weights, we **avoid computing correlation diagrams** that complicate classical NTK analysis, while maintaining rigor. This simplification enables us to derive explicit NTK recursions, characterize the spectral structure of the kernel matrix, and establish concentration bounds—all while providing insights into how the landscape geometry shapes optimization dynamics.

The loss landscape of neural networks plays a crucial role in determining trainability and generalization. Without RKHS advantage, the Laplace kernel's quick spectral decay prevents high-frequency learning [2, 9], while sharp minima can lead to poor generalization [10, 11]. The stepwise loss behavior observed in MMNNs [1]—where training exhibits long plateaus followed by sharp drops—suggests that the landscape undergoes significant transitions during optimization. By characterizing the NTK spectrum and its evolution, we provide a framework for understanding these landscape transitions and explaining how the low-rank structure enables the global convergence observed empirically.

1.1 Related work

The Neural Tangent Kernel (NTK) framework [2] characterizes infinitely wide neural networks trained by gradient descent as kernel methods with a fixed kernel. This framework has been extended to finite width [3], convolutional architectures [4], and depth [5, 6], establishing a comprehensive theoretical foundation for understanding wide network training. Recent work by [12] characterizes the eigenvalue distribution of the NTK in the quadratic scaling regime (width $N \sim n^2$) for single-layer networks, establishing deformed Marchenko–Pastur laws with spike-bulk structure. Their analysis requires careful tracking of correlation diagrams and moment calculations. We extend this spectral analysis to the two-layer case with random features, where the rank-r bottleneck and frozen feature weights fundamentally alter the kernel's spectral properties. A key technical advantage of our approach is that we **avoid computing correlation diagrams** by leveraging the decoupling properties of Fisher–Kibble distributions and the independence structure inherent in frozen random features, which simplifies the analysis while maintaining rigor. Parallel to this, random feature approximations [13, 14] provide a practical route to kernel approximation by sampling features from suitable distributions, with rigorous guarantees for generalization [15, 16] and approximation [17]. Notably, freezing random features ensures full support throughout training, so universal approximation persists at all finite times—a property stronger than classical kernel ridge assumptions.

Low-rank and factorized architectures [18, 19, 20] substantially reduce parameter counts in neural networks. However, the training dynamics of factorized weights can differ from unfactorized gradient descent due to implicit regularization effects. When only output weights are trained while feature weights remain frozen, the architecture avoids Riemannian geometry complications while achieving linear parameter scaling $O(rN)$, representing an extreme case of factorization that enables clean theoretical analysis.

Neural networks exhibit a well-documented spectral bias toward low frequencies [21, 22], where low-frequency components are learned before high-frequency ones. This bias has been quantified via NTK analysis [23] and Fourier methods [24], revealing how kernel structure shapes learning dynamics. The "deep equals shallow" result for ReLU kernels in the NTK regime [25] establishes that the RKHS induced by deep networks coincides with that of a shallow ReLU kernel, providing formal NTK spectrum decay rates. The same work characterizes the eigenvalue decay of rotation-invariant kernels on the sphere via endpoint expansions of the kernel $\kappa(\rho)$ near $\rho = \pm 1$, relating the smoothness properties of the kernel at the endpoints to the decay rate of spherical-harmonic eigenvalues. This connection between kernel smoothness and spectral decay provides a principled framework for understanding how RKHS structure, kernel spectra, and training behavior interact.

Multi-index approaches [26, 27, 28] learn functions of the form $f(x) = \sum_{j=1}^{r} g_j(\langle v_j, x\rangle)$ by discovering latent directions sequentially, with training displaying staircase dynamics featuring plateaus for inactive directions. In extensive-rank regimes $r \asymp d^{\beta}$, these works derive scaling laws [27] that connect rank, dimension, and sample complexity. Complementarily, dictionary learning [29, 30] and its neural analogues [31, 32] interpret networks as selecting sparse combinations from overcomplete bases, suggesting that training effectively performs implicit dictionary selection.

On the optimization side, gradient descent near sharp minima can be unstable [33], with edge-of-stability (EOS) phenomena causing loss oscillations when eigenvalues exceed $2/\eta$. Different optimizers exhibit distinct implicit biases [34], and in sharp regimes switching to SGD can improve stability. Neural scaling laws [35, 36] describe test loss as power laws in compute, data, and model size, while emergent capabilities [37] appear at critical scales. Universal approximation theorems [38, 39] and random-feature universality [14] ensure expressivity, and in three-layer frozen-feature settings, approximation power is maintained at *all* training times [40]. Loss landscape geometry [10, 11] significantly affects trainability; as training progresses, the landscape often convexifies [41], with Hessian spectrum evolution providing quantitative measures of this transition.

From a data-dependent scaling perspective, achieving and maintaining NTK behavior in fully-trainable deep networks can impose stringent width–data requirements, often quoted in the literature as polynomial $O(N^p)$ in the width N, with worst-case exponents p tied to depth and curvature [3, 5, 6]. To ensure the NTK has a positive smallest eigenvalue—crucial for optimization convergence—at least one layer must scale with the number of data samples n, while other layers can scale arbitrarily up to logarithmic factors [42]. This polynomial scaling can become prohibitive for deep networks, motivating architectures that achieve NTK-like behavior with more favorable computational requirements.

Our Contributions

Our contributions are: (i) an NTK recursion specialized to RF-LR together with finite-width corrections and concentration rates, (ii) a two-layer spectral theory yielding a signal–noise decomposition of the empirical kernel and a deformed MP law with a spike for the kernel matrix spectrum, and (iii) a principled argument—grounded in [25]—that RF-LR suffers no RKHS disadvantage relative to deep fully-trainable networks. These theoretical results are complemented by numerical experiments (omitted here) that verify NTK spectral concentration, staircase loss dynamics, and the scaling benefits of RF-LR at moderate widths.

More specifically:

- **NTK recursion** We derive the new NTK recursion expansion for RF-LR without bias/output scaling ($\sigma_A, \sigma_c, \beta = 1$), yielding clean expressions for two-layer and general L-layer networks.
- **No RKHS advantage.** In the NTK regime $f_t(x) \approx f_0(x) + \langle \nabla_\theta f_0(x),\, \theta_t - \theta_0 \rangle$ and $f_t \in \mathcal{H}_\Theta$ with $\|f\|^2_{\mathcal{H}_\Theta} = \langle f, \Theta^{-1} f \rangle_{L^2}$. RF-LR preserves universal approximation (frozen Gaussian $w^{(\ell)}$), the rank-r bottleneck acts as a spectral filter, and its NTK-RKHS has the same decay rate as an MLP's while using $O(rN)$ rather than $O(N^2)$ parameters.
- **Spectral theory (two-layer case).** We state and analyze Fisher–Kibble decoupling [7, 8], a signal–noise decomposition of the empirical NTK, and a deformed Marchenko–Pastur law (with spike) for the kernel matrix spectrum under standard random matrix scaling.

2 Network definition and EOC parameterization

2.1 RF-LR architecture

Let $\boldsymbol{h}^{(0)}(x) = x \in \mathbb{R}^{\boldsymbol{d}_0}$. For $\ell = 1, \ldots, L$,

$$\boldsymbol{h}^{(\ell)}(x) = \frac{1}{\sqrt{\boldsymbol{n}_\ell}} \sum_{j=1}^{\boldsymbol{n}_\ell} A_j^{(\ell)} \;\boldsymbol{\sigma}\left(w_j^{(\ell)\top} \boldsymbol{h}^{(\ell-1)}(x) + b_j^{(\ell)} \right) \;+\; c. \tag{1}$$

Training policy: $A^{(\ell)}$ and the scalar output bias c are trained; $w^{(\ell)}$ and activation biases $b^{(\ell)}$ are frozen random draws (i.i.d. Gaussian). The $1/\sqrt{\boldsymbol{n}_\ell}$ scaling (NTK parameterization) ensures a well-defined infinite-width limit. Low-rank bottlenecks (intermediate dimension $r \ll \boldsymbol{n}_\ell$) compress correlations, reduce parameter budget from $O(n^2)$ to $O(rn)$, and improve conditioning.

Training parameters, biases, and initialization. Initialization is

$$w_j^{(\ell)} \sim \mathcal{N}\big(0, I_{\boldsymbol{d}_{\ell-1}}/\boldsymbol{d}_{\ell-1}\big), \quad b_j^{(\ell)} \sim \mathcal{N}(0, \beta^2), \quad A_{ij}^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2), \quad c \sim \mathcal{N}(0, \sigma_c^2),$$

We use the *edge-of-chaos (EOC)* scaling for the weights so that $\mathrm{Cov}\big(w_j^{(\ell)}\big) = I_{\boldsymbol{d}_{\ell-1}}/\boldsymbol{d}_{\ell-1}$ [43]. The low-rank constraint is enforced as $\mathrm{rank}\big(A^{(\ell)}\big) \le r_\ell \ll \boldsymbol{n}_\ell$.

We include an activation bias $b^{(\ell)}$ and a scalar output bias c for completeness, but we avoid introducing extra scaling parameters for them. Output scaling can be absorbed into the learning rate; the mean-field and NTK structures remain qualitatively identical, and the Fourier-space interpretation and stepwise loss phenomenon depend on kernel geometry, not absolute scales.

With this setup in place, we now recall a minimal assumption on the activation that we will use throughout.

Activation homogeneity. We only assume a positively 1-homogeneous nonlinearity $\boldsymbol{\sigma}$ for Fisher–Kibble decoupling (see Lemma 9.1):

$$\boldsymbol{\sigma}\,(\alpha u) \;=\; \alpha\;\boldsymbol{\sigma}\,(u) \quad \text{for } \alpha \ge 0,\; u \in \mathbb{R}.$$

This yields the scaling relation $\boldsymbol{\sigma}\;(\alpha\, w^\top h) = \alpha\;\boldsymbol{\sigma}\;(w^\top h)$ and, away from zero, $\dot{\boldsymbol{\sigma}}(\alpha u) = \dot{\boldsymbol{\sigma}}(u)$.

2.2 Parameterization, signal propagation, and kernels

With the model and initialization fixed, we can now examine parameterization choices, signal propagation at initialization, and the kernel quantities driving our analysis. We begin by tracking the layer-wise variance at initialization.

For general weight variance $\Sigma_w = I_{d_{\ell-1}}/d_{\ell-1}$, define the per-layer variance $q^{(\ell)}$, by weight independance we derive it's recursion:

$$q^{(\ell)} = \sigma_c^2 + \frac{\sigma_A^2}{2} q^{(\ell-1)} \operatorname{Tr}(\Sigma_w),$$

This leads to the *(EOC)* condition:

$$\frac{\sigma_A^2}{2} \operatorname{Tr}(\Sigma_w) = 1.$$

it is chosen to prevent activations (forward) and gradients (backward) from exploding or vanishing as they propagate through the layers. By tuning the initialization so that the update satisfies this fixed point, both the signal and its derivative maintain stable variance at each layer. For example, under $\Sigma_w = I_{d_{\ell-1}}/d_{\ell-1}$, the EOC choice gives $\sigma_A = \sqrt{2}$ and typically $\sigma_c^2 = 1$.

With an abuse of notation, we denote by $\Sigma^{(\ell)}$ the base kernel and by $\dot{\Sigma}^{(\ell)}$ the derivative kernel. We emphasize that these are not the NNGP kernel and the derivative kernel, but the base kernel and the derivative kernel of the Gaussian process $\boldsymbol{h}^{(\ell)}$ conditional on $\boldsymbol{h}^{(\ell-1)}$. By rotation invariance of Gaussian weights, these kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \, \|x'\|)$. We will therefore write $\Sigma^{(\ell)}(\rho)$ and $\dot{\Sigma}^{(\ell)}(\rho)$. When no confusion arises (e.g., for $\ell = 1$), we will also abuse notation and write $\sigma(\rho)$ for the scalar kernel function; this σ denotes the kernel-as-a-function-of-ρ, not the activation $\boldsymbol{\sigma}$. We can now state the base and derivative kernels precisely.

Definition 2.1 (Base and derivative kernels)**.** In the sequential infinite-width limit, the layer-ℓ base kernel is

$$\Sigma^{(\ell)}(x, x') = \mathbb{E}_w \Big[\boldsymbol{\sigma}\left(w^\top \boldsymbol{h}^{(\ell-1)}(x)\right) \; \boldsymbol{\sigma}\left(w^\top \boldsymbol{h}^{(\ell-1)}(x')\right) \Big], \tag{2}$$

and the derivative kernel is

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \Big[\dot{\boldsymbol{\sigma}}\big(w^\top \boldsymbol{h}^{(\ell-1)}(x)\big) \, \dot{\boldsymbol{\sigma}}\big(w^\top \boldsymbol{h}^{(\ell-1)}(x')\big) \; \|w\|^2 \Big]. \tag{3}$$

For $\ell = 1$ and centered Gaussian w, $\Sigma^{(1)}$ is rotation-invariant with standard closed forms in terms of input cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$:

$$\Sigma^{(1)}(x, x') = \frac{\|x\| \|x'\|}{2\pi} \big((\pi - \theta) \cos\theta + \sin\theta \big), \quad \theta = \arccos(\rho).$$

Given these definitions, we can now state the following composition result.

Theorem 2.1 (Gaussian process composition)**.** *In the sequential infinite-width limit, the hidden states $\boldsymbol{h}^{(\ell)}$ form a composition of Gaussian processes:*

- *Conditionally on $\boldsymbol{h}^{(\ell-1)}$, for any finite collection of inputs $\{x_1, \ldots, x_m\}$, the vector $(\boldsymbol{h}^{(\ell)}(x_1), \ldots, \boldsymbol{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian.*
- *The conditional limiting Gaussian has mean zero and covariance matrix $[\Sigma^{(\ell)}(x_i, x_j)]_{i,j=1}^m$, where $\Sigma^{(\ell)}$ is the base kernel defined in Definition 2.1 and depends on $\boldsymbol{h}^{(\ell-1)}$*

Proof. See Appendix 8.2. □

Composition of Gaussian processes We highlight the main difference between the base kernel and the NNGP kernel arising in classical NTK analysis : Since $\boldsymbol{h}^{(\ell-1)}$ itself is a (conditional) Gaussian process, the overall distribution is a composition of Gaussian processes, not a single Gaussian process.

3 NTK recursion

We now state the main result characterizing the NTK recursion for MMNNs. This recursion governs how the kernel evolves through layers and determines the training dynamics in the infinite-width limit.

Theorem 3.1 (Infinite-width NTK recursion)**.** *Under NTK parameterization $1/\sqrt{\boldsymbol{n}_\ell}$ and a sequential limit $\boldsymbol{n}_1 \to \infty, \dots, \boldsymbol{n}_L \to \infty$ with $\boldsymbol{w}^{(\ell)} \sim \mathcal{N}(0, \mathbf{I}_{\boldsymbol{d}_{\ell-1}})$ i.i.d. frozen after initialization, the layer-ℓ NTK satisfies*

$$\mathbf{\Theta}^{(0)}(x, x') \;=\; 0, \tag{4}$$

$$\mathbf{\Theta}^{(\ell)}(x, x') \;=\; 1 + \mathbf{\Theta}^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') \;+\; \Sigma^{(\ell)}(x, x'), \quad \ell \geq 1, \tag{5}$$

For $\ell = 1$, $\Sigma^{(1)}$ is deterministic; for $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields whose fluctuations shrink with the rank r of bottlenecks.

Furthermore, the gradient flow dynamics in the NTK regime satisfy

$$\frac{d}{dt} f_t(x) = -\int \mathbf{\Theta}^{(L)}(x, x')\big(f_t(x') - y(x')\big) d\mu(x'),$$

where μ is the data distribution and y is the target function.

Proof: See Appendix 8.1, Proof 8.1.

No weights particle coupling We can now interpret the structure of the recursion. Similarly to the original MLPs NTK recursion [2], the additive term $\Sigma^{(\ell)}$ encodes fresh basis contribution at layer ℓ, while the multiplicative coupling $\mathbf{\Theta}^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}$ reflects the backward through layers via the derivative kernel, since there is no coupling between weights particles between 2 layers during training due to random features in the middle. Nonetheless, this decoupling that makes the training easier to tackled under the mean-field limit is effectively lost at the NTK limit and allows to factorize the derivative kernel.

We can now derive an explicit closed-form expression for the NTK at any depth by expanding the recursion.

Corollary 3.1 (General L-layer NTK; closed form)**.** *Under the assumptions of Theorem 3.1, for any $L \geq 1$, the NTK admits the explicit form:*

$$\mathbf{\Theta}^{(L)}(x, x') = \sum_{\ell=1}^{L} \left(\prod_{k=\ell+1}^{L} \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^{L} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{L} \dot{\Sigma}^{(k)}(x, x'), \tag{6}$$

where the first sum represents contributions from the bias terms +1 at each layer $\ell = 1, \dots, L$, each multiplied by all subsequent derivative kernels, and we use the convention that an empty product equals 1.

Remark 3.1. Equivalently, expanding the recursion $\mathbf{\Theta}^{(\ell)} = 1 + \mathbf{\Theta}^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)} + \Sigma^{(\ell)}$ yields a quadratic number of terms. For example, for $L = 3$:

$$\begin{aligned} \mathbf{\Theta}^{(3)}(x, x') = {} & \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \dot{\Sigma}^{(3)} + 1 \\ & + \Sigma^{(1)} \cdot \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \Sigma^{(2)} \cdot \dot{\Sigma}^{(3)} + \Sigma^{(3)}. \end{aligned} \tag{7}$$

The first three terms (from the first sum with $\ell = 1, 2, 3$) come from the +1 contributions at layers 1, 2, and 3, each multiplied by all subsequent derivative kernels. The last three terms (from the second sum with $\ell = 1, 2, 3$) come from the $\Sigma^{(\ell)}$ contributions, each multiplied by subsequent derivative kernels. In total, there are $2L = 6$ terms for $L = 3$.

Proof. By expanding the recursion in Theorem 3.1. See Appendix 8.4 for the detailed proof by induction. □

Structure at depth L. Iterating the recursion $\mathbf{\Theta}^{(\ell)} = 1 + \mathbf{\Theta}^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)} + \Sigma^{(\ell)}$ yields a quadratic number of terms. From the explicit form (6), we see two types of contributions:

- **Bias terms:** Each layer's +1 contribution multiplies through all subsequent derivative kernels, giving L terms of the form $\prod_{k=\ell+1}^{L} \dot{\Sigma}^{(k)}$ for $\ell = 1, \ldots, L$.
- **Fresh basis terms:** Each layer's $\Sigma^{(\ell)}$ contribution propagates through subsequent derivative kernels, giving L terms of the form $\Sigma^{(\ell)} \prod_{k=\ell+1}^{L} \dot{\Sigma}^{(k)}$ for $\ell = 1, \ldots, L$.

In total, the expansion contains $2L$ terms, each representing a different path through the network architecture. This linear growth in the number of terms (with respect to depth) becomes quadratic when considering all combinations of terms in finite-width corrections, making explicit computation and analysis very difficult to handle for deep networks.

We now specialize to the two-layer case, where we can obtain a particularly compact form under a homogeneity assumption on the activation function.

Corollary 3.2 (Two-layer NTK and homogeneous activation compact form)**.** *Under the assumptions of Theorem 3.1 with $L = 2$,*

$$\mathbf{\Theta}^{(1)}(x, x') \;=\; 1 + \Sigma^{(1)}(x, x'),$$

$$\mathbf{\Theta}^{(2)}(x, x') \;=\; 1 + \mathbf{\Theta}^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') \;+\; \Sigma^{(2)}(x, x'),$$

Proof. Immediate from Theorem 3.1. □

We can now state a more compact representation under a homogeneity assumption.

3.1 Microscopic distributions: Fisher–Kibble decoupling

Lemma 3.1 (Fisher–Kibble decoupling)**.** *Let x, y be input vectors and let x_1, y_1 denote their rank-r random projections. Define the empirical correlation $\rho_1 = \cos\angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- *ρ_1 follows Fisher's correlation distribution [7, 8], centered at the true correlation ρ.*
- *(u, v) follow Kibble's (1941) bivariate Gamma (chi-square) law.*
- *Angular and radial parts are independent: $p(\rho_1, u, v) = p_{Fisher}(\rho_1)\, p_{Kibble}(u, v)$.*

Proof: See Appendix 9.3, Proof 9.3.

Homogeneity assumption enabling decoupling. The independence of angular and radial components relies on the *positive 1-homogeneity* of the ReLU activation: $\boldsymbol{\sigma}(\alpha z) = \alpha\, \boldsymbol{\sigma}(z)$ for $\alpha > 0$. This property ensures that the base kernel factorizes as

$$\Sigma^{(1)}(x, x') = \|x\|\, \|x'\| \cdot \tilde{\Sigma}^{(1)}(\rho),$$

where $\tilde{\Sigma}^{(1)}(\rho)$ depends only on the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$, enabling the separation of radial norms from angular correlations. For isotropic Gaussian projections, rotational invariance guarantees that the empirical correlation ρ_1 and the norms $(\|x_1\|, \|y_1\|)$ are statistically independent, yielding the factorization stated in the lemma.

This decoupling result enables us to derive the asymptotic distribution of the empirical kernel. We can state the following central limit theorem.

In the two-layer low-rank setting, denote

$$\rho_1 \;=\; \cos\angle(\boldsymbol{h}^{(1)}(x), \boldsymbol{h}^{(1)}(x')), \qquad w_r \;=\; \frac{\|\boldsymbol{h}^{(1)}(x)\|\, \|\boldsymbol{h}^{(1)}(x')\|}{r},$$

then *both ρ_1 and w_r are random variables* induced by the randomness of the features and the rank- r projection; their values fluctuate with the initialization and with r. The NTK admits the compact representation

$$\mathbf{\Theta}^{(2)}(x, y) \;=\; \mathbf{\Theta}^{(1)}(\rho_1)\left(1 - \frac{\arccos(\rho_1)}{\pi}\right) \;+\; w_r\, \Sigma^{(1)}(\rho_1) \;+\; 1$$

with $\Sigma^{(1)}(u) = \frac{1}{\pi}\big(\sqrt{1-u^2} + u(1 - \arccos u)\big)$.

Remark (Fisher–Kibble decoupling and independence). We can now characterize the statistical properties of the empirical correlation and norms. By rotational invariance, the empirical correlation and squared norms follow Fisher–Kibble decoupling [7, 8]:

$$\rho_1 \sim \text{Fisher}(\rho, r), \qquad (\|\boldsymbol{h}^{(1)}(x)\|^2, \|\boldsymbol{h}^{(1)}(x')\|^2) \sim \text{Kibble}(r, \rho),$$

We now state the explicit forms of these distributions. The **Fisher distribution density** [7] is

$$\boxed{p(\rho_1 \mid \rho, r) = \frac{(r-2)\,\Gamma(r-1)\,(1-\rho^2)^{\frac{r-1}{2}}\,(1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(1-\rho\rho_1)^{r-\frac{3}{2}}}\,{}_2F_1\Big(\tfrac{1}{2}, \tfrac{1}{2};\, r-\tfrac{1}{2};\, \tfrac{1+\rho\rho_1}{2}\Big),}$$

for $r > 2$. For large r, the inverse Gudermann function applied to the complementary angle follows a normal distribution:

$$\mathrm{gd}^{-1}\Big(\frac{\pi}{2} - \arccos(\rho_1)\Big) \;\approx\; \mathcal{N}\Big(\operatorname{arctanh}(\rho),\; \frac{1}{r-3}\Big)$$

where $\rho_1 = \cos(\theta_1)$ is the empirical correlation and θ_1 is the angle between the projected vectors. In particular, this implies $\mathrm{Var}(\rho_1) = O(1/r)$.

Similarly, **Kibble's [44] bivariate chi-square law** for $(u, v) = (\|\boldsymbol{h}^{(1)}(x)\|^2, \|\boldsymbol{h}^{(1)}(x')\|^2)$ has density

$$\boxed{f(u,v) = \frac{(uv)^{\frac{r/2-1}{2}}\,\exp\Big(-\frac{u+v}{2(1-\rho^2)}\Big)}{\Gamma(r/2)\,\big(2(1-\rho^2)\big)^{\frac{r}{2}+1}\,\rho^{\frac{r/2-1}{2}}}\,I_{\frac{r}{2}-1}\Big(\frac{\rho\sqrt{uv}}{1-\rho^2}\Big), \qquad u, v \geq 0,}$$

where I_ν is the modified Bessel function of the first kind. Defining $w_r = \sqrt{uv}/r$, concentration results (see Theorem 4.1) imply $\mathbb{E}[w_r] \to 1$ and $\mathrm{Var}(w_r) = O(1/r)$.

The moments of this distribution are given by:

$$\mathbb{E}[u] = \mathbb{E}[v] = r, \quad \mathrm{Var}(u) = \mathrm{Var}(v) = 2r, \quad \mathrm{Cov}(u, v) = 2r\,\rho^2.$$

The derivation of these distributions from Gaussian projections and the independence property are given in Appendix 9.3.

4 No RKHS advantage

In the NTK regime, training dynamics linearize: $f_t(x) \approx f_0(x) + \langle \nabla_\theta f_0(x), \theta_t - \theta_0\rangle$. The learned function stays in the RKHS induced by Θ, with norm:

$$\|f\|^2_{\mathcal{H}_\Theta} = \langle f, \Theta^{-1} f\rangle_{L^2}.$$

For RF-LR with frozen random features, the RKHS is characterized as follows. Universal approximation holds due to full support of $w^{(\ell)}$ (frozen Gaussian draws), so any function in the Barron space [45] can be approximated with error $O(1/\sqrt{N})$. So the RKHS is the same as for MLPs NTK analysis shows weights are nearly frozen through training. We tackle directly this remark by freezing the weights instead of performing weights-deviation analysis on MLPs. This also removes the EOC condition on the bias.

We can now state a result highly similar to [25]: even for in the extensive rank regime, RF-LR NTK under the NTK perspective, deeper architectures do not yield a larger RKHS than shallow networks.

4.1 Rank-driven concentration in RKHS (Gaussian norms product)

Having established that low rank and depth confers no RKHS advantage, we now turn to quantifying how the low-rank structure controls kernel fluctuations. We can show that the radial component of the RF-LR NTK concentrates exponentially fast in the rank r, yielding high-dimensional control of RKHS fluctuations. To state this result, we first require a homogeneity assumption on the activation function.

Assumption 4.1 (Homogeneous activation)**.** The activation function σ is positively homogeneous: $\sigma(\alpha z) = \alpha\sigma(z)$ for all $\alpha \geq 0$ and $z \in \mathbb{R}$. In particular, ReLU satisfies this property.

We can now state our main concentration result.

Theorem 4.1 (Concentration in rank for homogeneous activation)**.** *Under homogeneous activation assumption (Assumption 2.1), let $x_1, y_1 \in \mathbb{R}^r$ be the rank-r Gaussian projections of inputs x, y, with arbitrary fixed correlation $\rho \in [-1, 1]$. Define*

$$W = \frac{\|x_1\| \, \|y_1\|}{r}.$$

Then for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\,\epsilon^2}{8}\right).$$

Moreover, for larger deviations $\epsilon \geq 1$, there exist absolute constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 \exp(-c_2\, r\, \epsilon)\,.$$

In particular, the radial component of the RF-LR NTK concentrates exponentially fast in r, yielding high-dimensional control of RKHS fluctuations.

We now provide an intuitive explanation of the proof strategy.

Proof intuition. Set $U = \|x_1\|/\sqrt{r}$ and $V = \|y_1\|/\sqrt{r}$. U and V concentrate around 1 with sub-Gaussian tails $\exp(-r\delta^2)$. Using $|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|$ and a union bound with $\delta = \epsilon/2$ yields the stated $4\exp(-r\epsilon^2/8)$ bound. The full proof is given in Appendix 9.1.

Remark 4.1. The result avoids direct manipulation of Kibble's bivariate chi-square density, relying instead on Gaussian isoperimetry for norms and a simple algebraic bound for products.

4.2 Two-layer NTK concentration

We can now apply this concentration result to obtain bounds for the two-layer NTK.

Corollary 4.1 (Concentration bound for two-layer NTK)**.** *Under Assumption 2.1, let $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ be the empirical two-layer NTK, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\|\|y_1\|/r$, with $x_1, y_1 \in \mathbb{R}^r$ being the rank-r Gaussian projections of inputs x, y with population correlation $\rho \in [-1, 1]$. Let $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ be the deterministic kernel limit.*

Then for any $\epsilon \in (0, 1)$, there exist constants $C_1, C_2 > 0$ depending on ρ and the kernel smoothness such that

$$\mathbb{P}\left(\left|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)\right| \geq \epsilon\right) \leq C_1 \exp\left(-C_2\, r\, \epsilon^2\right).$$

Proof: See Appendix 9.2.

Remark 4.2 (Critical cancellation of $1/r$ corrections). While the general concentration rate is $O_p(1/\sqrt{r})$ via Fisher's CLT [7, 8], the randomness induced from the gaussian process low rank output remove the $1/r$ correction term in $\Sigma^{(2)}$ that would normally appear through the radial component $\hat{w}_r$. This cancellation is critical: it is precisely the path where finite-width corrections $1/N$ analysis [46] and Feynman diagram decay come into play, proving that the fluctuation rate is $1/r$ rather than $1/\sqrt{r}$.

Remark 4.3 (Extension to general L layers). The extension of this concentration bound to general depth L layers under Assumption 2.1 requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure. The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\|\|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. This analysis is deferred to a future version.

4.3 Mean NTK over Fisher and Kibble distributions

Having established concentration results, we now turn to the mean NTK. The key observation is that the empirical NTK depends on random variables that follow Fisher [7, 8] and Kibble distributions. The mean NTK is obtained by taking expectations over these distributions.

For the two-layer case, the empirical NTK is:

$$\hat{\Theta}_r^{(2)}(x, x') = \Theta^{(1)}(\rho) \cdot \left(1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right) + \frac{1}{r} \cdot \Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\|\|y_1\| + 1,$$

where $\hat{\rho}_r$ follows Fisher's distribution [7, 8] (centered at ρ) and $\|x_1\|\|y_1\|$ follows Kibble's distribution. The mean NTK is therefore:

$$\tilde{\mathbf{\Theta}}^{(2)}(\rho) = \mathbb{E}_{\text{Fisher,Kibble}}\left[\hat{\Theta}_r^{(2)}(x, x')\right] = \Theta^{(1)}(\rho) \cdot \mathbb{E}\left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right] + \frac{1}{r} \cdot \mathbb{E}\left[\Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\|\|y_1\|\right] + 1,$$

where the expectations are taken over the Fisher distribution of $\hat{\rho}_r$ and the Kibble distribution of $(\|x_1\|^2, \|y_1\|^2)$.

Corollary 4.2 (Mean NTK over Fisher–Kibble has same RKHS)**.** *Under Assumption 2.1 and the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the mean NTK $\tilde{\mathbf{\Theta}}^{(L)}$ (obtained by taking expectations over Fisher and Kibble distributions at each layer) is a zonal kernel on $\mathbb{S}^{d-1}$ that induces the same RKHS as the shallow (two-layer) ReLU kernel. In particular, the mean NTK RKHS coincides with the unconditional RKHS as sets with equivalent norms.*

For the two-layer case ($L = 2$), the mean NTK is:

$$\tilde{\mathbf{\Theta}}^{(2)}(\rho) = \Theta^{(1)}(\rho) \cdot \mathbb{E}_{\hat{\rho}_r \sim Fisher(\rho, r)}\left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right] + \frac{1}{r} \cdot \mathbb{E}_{\hat{\rho}_r, \|x_1\|, \|y_1\|}\left[\Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\|\|y_1\|\right] + 1,$$

where the expectations are over Fisher [7, 8] and Kibble distributions. This yields a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$, but preserves the same endpoint behavior, ensuring RKHS equivalence.

Intuition and technical challenges. This corollary is highly non-trivial because it requires computing expectations of $\arccos(\hat{\rho}_r)$ where $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$ [7, 8], which involves integrating over Fisher's full distribution density near the boundary $\rho = 1$. The key technical challenge is analyzing the Puiseux expansion of the mean NTK near $\rho = 1$, where both $\arccos(\hat{\rho}_r)$ and Fisher's density exhibit singular behavior.

The proof (Appendix 9.4) computes the integral:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] \;=\; \int_{-1}^{1} \arccos(u)\; \frac{(r-2)\,\Gamma(r-1)\,\left(1-(1-t)^2\right)^{\frac{r-1}{2}}\,(1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma\!\left(r-\frac{1}{2}\right)\,\left(1-(1-t)\,u\right)^{r-\frac{3}{2}}}\; {}_2F_1\!\left(\tfrac{1}{2},\, \tfrac{1}{2};\, r-\tfrac{1}{2};\, \tfrac{1+(1-t)\,u}{2}\right)\, du,$$

Near this boundary, we rely on asymptotic expansions of hypergeometric functions near their argument $z \to 1^-$ (see [47] and [48]).

The crucial insight is that while the mean NTK has a different Puiseux expansion than the deterministic kernel $K_\infty(\rho)$, the leading-order $t^{1/2}$ term (which determines spectral decay and RKHS structure) is preserved. This preservation of endpoint behavior under Fisher–Kibble expectations is what ensures RKHS equivalence, despite the different higher-order corrections that appear in the mean kernel.

4.4 Spectral decay and Puiseux expansions

We now apply the spectral decay theory of [25] to characterize the RKHS of the RF-LR NTK via endpoint expansions. The base kernel $\Sigma^{(1)}$ and derivative kernel $\dot{\Sigma}^{(1)}$ are the fundamental building blocks for the NTK recursion.

As a reminder, we have puiseux closed forms for classical MLPs NTK :

$$\Sigma^{(1)}(1-t) = 1 - \frac{2}{\pi} t^{1/2} + O(t^{3/2}), \qquad \dot{\Sigma}^{(1)}(1-t) = \frac{d}{2} - \frac{d}{2\pi} t^{1/2} + O(t^{3/2}).$$

From the proof, near $\rho = 1$ (for $t \geq 0$), the mean two-layer NTK admits the Puiseux expansion:

$$\tilde{\mathbf{\Theta}}^{(2)}(1-t) = 1 - \frac{2}{\pi} t^{1/2} + O(t^{3/2}) + O(r^{-1} t^{1/2}),$$

where the leading non-polynomial term has exponent $\nu = 1/2$ with coefficient $c_1 = -2/\pi$. Similarly, near $\rho = -1$, by rotational symmetry and homogeneity of ReLU, the expansion has the same structure with $c_{-1} = c_1 = -2/\pi$ (see [25]).

We can now apply Theorem 1 of [25] to obtain the spectral decay characterization:

Theorem 4.2 (Spectral decay for mean RF-LR NTK with final constant)**.** *Let $\kappa(\rho) = \tilde{\mathbf{\Theta}}^{(2)}(\rho)$ be the mean NTK on $\mathbb{S}^{d-1}$. From the Puiseux expansion in Appendix (see the explicit constant at the end of Appendix 9.4), κ admits, for $t \geq 0$,*

$$\kappa(1-t) = p_1(t) + c_1(r)\, t^{1/2} + o(t^{1/2}), \qquad \kappa(-1+t) = p_{-1}(t) + c_{-1}(r)\, t^{1/2} + o(t^{1/2}),$$

with exponent $\nu = 1/2$. For ReLU and normalized inputs,

$$c_1(r) \;=\; -\left[\frac{2}{\pi} \;+\; \frac{\sqrt{2}}{2\pi}\, I(r)\right], \qquad I(r) \;=\; \frac{(r-2)\, 2^{r-\frac{5}{2}}\, \Gamma\!\left(\frac{r-1}{2}\right) \Gamma\!\left(\frac{r}{2}-1\right)}{\sqrt{2\pi}\, \Gamma(r-1)} \cdot \frac{\Gamma\!\left(r-\frac{1}{2}\right) \Gamma\!\left(r-\frac{3}{2}\right)}{\Gamma(r-1)^2},$$

and, by symmetry of the ReLU kernel on $\mathbb{S}^{d-1}$, $c_{-1}(r) = c_1(r)$.

Then by Theorem 1 of [25], the eigenvalues μ_k (in the spherical harmonics basis) satisfy:

$$\mu_k \sim C(d,\nu)\, k^{-d-2\nu+1} = C(d)\, k^{-d-2(1/2)+1} = C(d)\, k^{-d},$$

where $C(d) > 0$ depends only on d. Since $c_1(r) = c_{-1}(r)$, we have $|c_1(r)| = |c_{-1}(r)|$, hence:

- *For even k: $\mu_k \sim \big(c_1(r) + c_{-1}(r)\big)\, C(d)\, k^{-d} = 2\, c_1(r)\, C(d)\, k^{-d}$.*
- *For odd k: since $c_1(r) - c_{-1}(r) = 0$, $\mu_k = o(k^{-d})$ (faster-than-polynomial in this parity).*

Therefore, the dominant eigenvalue decay rate is k^{-d}, determined by the even-parity eigenvalues.

Corollary 4.3 (No advantage in RKHS)**.** *Under the RF-LR setting with ReLU nonlinearity, isotropic random features on the sphere $\mathbb{S}^{d-1}$, and in the kernel (NTK) regime, the depth-L NTK induces the* same *RKHS as the shallow (two-layer) ReLU kernel. In particular, the corresponding RKHSs coincide as sets with equivalent norms. This is a direct application of Theorem 1 in [25].*

Remark 4.4. This theorem and corollary are direct applications of [25] to the RF-LR setting. Theorem 4.2 supplies explicit rates for approximation and generalization in the RF-LR NTK RKHS and, together with the explicit kernel formulas above, justifies that deeper NTKs do not enlarge the RKHS beyond the shallow ReLU case (Corollary 4.3).

5 RF-LR Gram Marchenko–Pastur spectrum for two-layer case

Now that we have established RKHS properties to clarify that random features hold frozen through training has same effect as letting them free to move, in a more applicative way we give NTK gram matrix spectrum asymptotic behavior. We hold strong data assumptions but could release them in future work.

5.1 General setup and scaling

Assumption 5.1 (Random-matrix scaling)**.** As the number of data points $n \to \infty$:

- Hidden widths scale as $N \sim n^2$ (infinite-width linearization regime).

- Rank scales as $r \asymp n$ with ratio $\gamma_{\text{ratio}} \equiv \lim_{n\to\infty} \frac{n}{r} \in (0,\infty)$ (Marchenko–Pastur regime).
- Input dimension scales linearly with rank: $d \asymp r$ (so $d \asymp n$ as well).

We now introduce the random-matrix theory conditions that are necessary for the asymptotic analysis. Those strong assumptions can be removed in future work using techniques from [12]

Assumption 5.2 (RMT Gaussian data)**.** Following the framework of [49], we impose the following conditions as $d \to \infty$:

(d) **Data generation:** Input vectors satisfy $X_i = \Sigma_d^{1/2} Y_i$, where $\Sigma_d^{1/2}$ is the positive semi-definite square root of Σ_d.

(e) **Entrywise conditions:** The entries of Y_i, a d-dimensional random vector, are i.i.d. Denoting by $Y_i^{(k)}$ the k-th entry of Y_i, we assume $\mathbb{E}[Y_i^{(k)}] = 0$, $\mathrm{var}(Y_i^{(k)}) = 1$, and $\mathbb{E}[|Y_i^{(k)}|^{4+\varepsilon}] < \infty$ for some $\varepsilon > 0$ (i.e., Y_i has $4+\varepsilon$ absolute moments).

(b) **Covariance structure:** Let Σ_d be the $d \times d$ covariance matrix of the input distribution. Then Σ_d is positive semi-definite and its operator norm $\|\Sigma_d\|_2 = \sigma_1(\Sigma_d)$ remains bounded in d, that is, there exists $K > 0$ such that $\sigma_1(\Sigma_d) \le K$ for all d.

(c) **Trace normalization:** Σ_d/d has a finite limit, that is, there exists $\ell \in \mathbb{R}$ such that $\lim_{d\to\infty} \mathrm{trace}(\Sigma_d)/d = \ell$.

With these scaling assumptions in place, we can now analyze the microscopic structure of the kernel. The key insight is that the randomness in the low-rank projections decomposes into independent angular and radial components.

To apply [49], the kernel function $K_\infty(\rho) = \Theta^{(1)}(\rho)\big(1 - \arccos(\rho)/\pi\big) + \Sigma^{(1)}(\rho) + 1$ must be C^1 in a neighborhood of $\ell = \lim_{d\to\infty} \mathrm{trace}(\Sigma_d)/d$ and C^3 in a neighborhood of 0. This holds trivially since $\arccos(\rho)$ is C^∞ on $(-1,1)$ and the ReLU kernels $\Theta^{(1)}(\rho)$ and $\Sigma^{(1)}(\rho)$ have the required smoothness properties.

5.2 Signal–noise decomposition of the empirical NTK

Theorem 5.1 (Asymptotic normality of the low-rank kernel)**.** *Let $x, y \in \mathbb{R}^r$ be rank-r Gaussian projections with population correlation ρ. Define*

$$\Psi(u,w) \;=\; \Theta^{(1)}(u)\left(1 - \frac{\arccos(u)}{\pi}\right) \;+\; w\,\Sigma^{(1)}(u) \;+\; 1,$$

and the empirical kernel $\hat\Theta_r^{(2)} \;=\; \Psi(\hat\rho_r, \hat w_r)$, where $\hat\rho_r$ is the sample correlation and $\hat w_r = \|x\|\,\|y\|/r$. Under Assumption 5.1, as $r \to \infty$,

$$\hat\Theta_r^{(2)} \;=\; \Theta_{\text{obs}}^{(2)}(\rho) \;\xrightarrow{d}\; K_\infty(\rho) \;+\; \frac{1}{\sqrt{r}}\,\mathcal{G}(\rho) + o(r^{-1/2}),$$

where $K_\infty(\rho) = \mathbb{E}[\Theta_{\text{obs}}^{(2)}(\rho)]$ is the deterministic limit and $\mathcal{G}(\rho) \sim \mathcal{N}\big(0, \sigma_{\text{tot}}^2(\rho)\big)$ is a centered Gaussian field with variance

$$\sigma_{\text{tot}}^2(\rho) \;=\; \mathrm{Var}(\mathcal{G}(\rho)) \;=\; \underbrace{\big[\partial_\rho K_\infty(\rho)\,(1-\rho^2)\big]^2}_{\textit{Fisher (angle) [7, 8]}} \;+\; \underbrace{\big[\Sigma^{(1)}(\rho)\,\sqrt{1+\rho^2}\big]^2}_{\textit{Kibble (norms)}}.$$

Proof intuition. By Fisher–Kibble decoupling (Lemma 9.1) [7, 8], the angular $\hat\rho_r$ and radial $\hat w_r$ components are independent. A joint CLT gives $\sqrt{r}\,(\hat\rho_r - \rho,\ \hat w_r - 1) \xrightarrow{d} \mathcal{N}(\mathbf{0},\ \mathrm{diag}((1-\rho^2)^2,\ 1+\rho^2))$. Applying the Delta method to Ψ at $(\rho, 1)$ with gradient $(\partial_\rho K_\infty(\rho),\ \Sigma^{(1)}(\rho))$ yields the stated variance (no cross term by independence). The limit is a deterministic kernel plus $O_p(r^{-1/2})$ Gaussian fluctuation.

Proof: See Appendix 10.1.

Having established the entrywise asymptotic distribution, we now turn to the macroscopic spectral properties of the kernel matrix. The eigenvalue distribution follows a deformed Marchenko–Pastur law with a characteristic spike-bulk structure.

5.3 Macroscopic spectrum: deformed Marchenko–Pastur law

Recent work by [12] establishes the eigenvalue distribution of the NTK in the quadratic scaling regime ($N \sim n^2$) for single-layer networks, showing convergence to a deformed Marchenko–Pastur law with a spike-bulk structure. We extend this analysis to the two-layer case with random features and rank-r bottlenecks, where the frozen feature weights and low-rank structure modify both the spike location and bulk support.

Theorem 5.2 (Spike + bulk spectral limit)**.** *Let* $\mathbf{M} = [\Theta^{(2)}_{ij}]_{i,j=1}^{n}$. *Under Assumptions 5.1 and 5.2, as* $n \to \infty$*, the empirical spectrum of* $\mathbf{M}$ *converges to a deformed Marchenko–Pastur law with:*

- *A rank-one outlier (spike) of order* $O(n)$*:* $\lambda_{\mathrm{spike}} \approx n\,\alpha$*, with* $\alpha = K_\infty(0)$*.*
- *A continuous bulk of order* $O(1)$ *supported on*

$$\Big[\ \gamma + \beta\,(1 - \sqrt{\gamma_{\mathrm{ratio}}})^2,\ \ \gamma + \beta\,(1 + \sqrt{\gamma_{\mathrm{ratio}}})^2\ \Big],$$

where $\beta = K'_\infty(0)$ *and* $\gamma = K_\infty(\ell) - \alpha - \ell\,\beta$ *(diagonal shift;* ℓ *denotes the kernel's diagonal argument).*

Proof: See Appendix 10.2 (via [49]). The extension to two layers with random features follows the framework of [12] but incorporates the additional randomness from rank-r projections and frozen feature weights.

We can summarize the two-layer results as follows.

Two-layer case summary. Specializing to two layers with ReLU features, the deterministic kernel limit is $K_\infty(\rho) = \Theta^{(1)}(\rho)\big(1 - \arccos(\rho)/\pi\big) + \Sigma^{(1)}(\rho) + 1$. At the dataset level, the corresponding Gram matrix exhibits a rank-one "spike" of size $\lambda_{\mathrm{spike}} \approx n\,K_\infty(0)$, superimposed on a deformed Marchenko–Pastur bulk. The bulk support is the interval given by Theorem 5.2, with slope $\beta = K'_\infty(0)$ and diagonal shift γ determined by the kernel's on-diagonal value.

6 Conclusion

This paper establishes the Neural Tangent Kernel (NTK) theory for low-rank random feature architectures (RF-LR/MMNN), demonstrating that these models provide a **clean and advantageous route** to the kernel regime with improved theoretical guarantees. **Our main result establishes that low-rank random features achieve the same RKHS expressivity as dense networks while requiring only a linear parameter budget $O(rN)$ instead of $O(N^2)$, with exponential concentration $O(1/r^2)$ faster than classical NTK theory predicts**.

Our primary contribution shows that there is **no RKHS disadvantage** for low-rank random features relative to standard fully-connected networks: under homogeneous activation and isotropic random features on the sphere $\mathbb{S}^{d-1}$, the depth-L NTK induces the **same RKHS** as the shallow (two-layer) ReLU kernel (Corollary 4.3). The induced RKHSs coincide as sets with equivalent norms, so expressivity is preserved. More importantly, the architecture enables the **kernel regime at a linear parameter budget** $O(rN)$ rather than the quadratic $O(N^2)$ of dense layers, where $r \ll N$ is the rank of the low-rank bottleneck and N is the width. This **linear-width scaling** lowers the threshold at which NTK predictions become accurate, enabling theoretical guarantees at smaller model sizes.

Beyond expressivity equivalence, we establish **exponential concentration properties** in the rank r. Under the homogeneity assumption, the radial component $W = \|x_1\|\|y_1\|/r$ concentrates exponentially fast: for any $\epsilon \in (0, 1)$, we have $\mathbb{P}(|W - 1| \geq \epsilon) \leq 4\exp(-r\epsilon^2/8)$ (Theorem 4.1). This yields exponential concentration for the two-layer NTK around its deterministic limit $K_\infty(\rho)$ (Corollary 4.1). The concentration relies on Fisher–Kibble decoupling (Lemma 9.1) [7, 8]: the angular component ρ_1 follows Fisher's correlation distribution, the radial norms follow Kibble's bivariate chi-square law, and these components are statistically independent due to rotational invariance.

The spectral analysis at initialization reveals a **bulk–outlier structure** consistent with random matrix theory. Under the scaling assumptions (Assumption 5.1), the NTK Gram matrix exhibits a **single dominant outlier** (spike) of order $O(n)$ at $\lambda_{\text{spike}} \approx n \cdot K_\infty(0)$, superimposed on a deformed Marchenko–Pastur bulk supported on an interval determined by the kernel's derivative at $\rho = 0$ and its diagonal value (Theorem 5.2). The asymptotic normality of the empirical kernel follows from a joint central limit theorem: $\sqrt{r}(\hat{\rho}_r - \rho, \hat{w}_r - 1) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \operatorname{diag}((1-\rho^2)^2, 1+\rho^2))$ (Theorem 5.1), enabling precise characterization of kernel fluctuations.

We develop the **NTK recursion** for arbitrary depth L: $\mathbf{\Theta}^{(\ell)}(x, x') = 1 + \mathbf{\Theta}^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') + \Sigma^{(\ell)}(x, x')$ (Theorem 3.1), which admits an explicit closed-form expansion with $2L$ terms representing different paths through the network (Corollary 3.1). For the two-layer case, we obtain a particularly compact representation in terms of the empirical correlation ρ_1 and norm product w_r, enabling tractable analysis. The mean NTK $\tilde{\mathbf{\Theta}}^{(L)}$, defined via nested conditional expectations over the Gaussian processes $h^{(\ell)}$, preserves the same RKHS structure (Corollary 4.2), and admits an expansion $\tilde{\mathbf{\Theta}}^{(L)}(\rho) = K_\infty(\rho) + \sum_{\ell=2}^{L} O(r^{-\ell})$ where each $O(r^{-\ell})$ term aggregates cross-layer contributions indexed by layer pairs (i, j) with $1 \leq i \leq j \leq \ell$.

The architectural design trains only output weights $A^{(\ell)}$ while freezing input weights $w^{(\ell)}$ as random Gaussian draws. This decoupling eliminates weight-particle coupling between layers, simplifying the mean-field analysis while preserving the NTK structure through the derivative kernel factorization. The edge-of-chaos (EOC) parameterization ensures stable signal propagation, with $\sigma_A^2 \cdot \operatorname{Tr}(\Sigma_w)/2 = 1$ preventing activations and gradients from exploding or vanishing.

In summary, viewed strictly through the NTK lens, low-rank random features provide a **clean and advantageous route** to the kernel regime: **the same RKHS** as dense networks (Corollary 4.3), **entry at linear budgets** $O(rN)$ versus $O(N^2)$, and **exponential concentration** in rank r (Theorem 4.1, Corollary 4.1). The **bulk–outlier spectral picture** (Theorem 5.2) illuminates initialization-time conditioning, and the **Fisher–Kibble decoupling** (Lemma 9.1) [7, 8] enables tractable analysis without computing correlation diagrams. These results establish a solid theoretical foundation for understanding RF-LR networks in the lazy training regime.

7 Discussion

The theoretical framework established in this work opens several promising directions for deepening our understanding of low-rank random feature architectures. While we have demonstrated fundamental advantages—linear parameter scaling, exponential concentration, and tractable spectral structure—much remains to be explored. The pathway forward is particularly exciting because low-rank random features offer a **unique combination of theoretical tractability and practical efficiency** that positions them as an ideal testbed for developing comprehensive theories of neural network behavior.

A distinguishing feature of our analysis is that we **avoid computing correlation diagrams** altogether by exploiting the independence structure of frozen random features and Fisher–Kibble decoupling [7, 8]. This simplification suggests that future generalizations can proceed with **fewer assumptions** while maintaining analytical control, following the direction pioneered by [12]. We are optimistic that this approach will enable extensions to more complex scenarios while preserving the computational advantages that make low-rank random features attractive.

The finite-width corrections framework we introduce—organized around the $1/r$ expansion $\tilde{\mathbf{\Theta}}^{(L)}(\rho) = K_\infty(\rho) + \sum_{\ell=2}^{L} O(r^{-\ell})$—provides a natural starting point for a systematic program aimed at elucidating the **finite-width loss landscape** under the NTK perspective. Each $O(r^{-\ell})$ term aggregates contributions from layer pairs (i, j) with $1 \le i \le j \le \ell$, arising from cross-layer interactions that are reminiscent of *ResNet*-style additive path contributions. These paths capture how Jacobian metrics and base kernels couple across layers, with each contribution bounded by $C_{\ell,i,j}\, r^{-\ell}$. A precise enumeration of these coefficients and potential closed-form resummations along dominant path families represents a concrete next step that would yield explicit finite-width formulas and deeper insights into how depth and rank interact.

On the theoretical front, several important questions remain open and constitute an immediate research agenda. The extension of the exponential concentration bounds to general depth L layers requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure (Remark after Corollary 4.1). The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\|\|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. A **non-Gaussian process propagation analysis** distinct from NNGP fits would provide complementary insights beyond the kernel regime, potentially revealing how feature learning emerges from deviations from Gaussianity. An **explicit bulk formula** with normalization scaled by r^2 would complete the spectral picture and enable direct comparisons with Marchenko–Pastur predictions. Relaxing the strong assumptions in Assumptions 5.1 and 5.2 (Gaussian data, specific scaling relationships) using techniques from [12] would extend the applicability of our spectral results to more realistic data distributions.

Additional theoretical extensions hold particular promise. The current analysis focuses on the two-layer case for tractability; extending to arbitrary depth L while maintaining explicit formulas remains an open challenge. Investigating how the edge-of-chaos (EOC) parameterization affects finite-width corrections and the Nr scaling mentioned in the architecture definition would provide concrete quantitative bounds. Understanding how the curse of dimensionality impacts normalization choices (unit norm weights vs. biases) depending on data geometry (hypercube vs. spherical) would provide principled guidance for practitioners.

The experimental validation of our theoretical predictions constitutes an equally important and concrete next step. We envision a comprehensive program that includes **confirming lazy-kernel predictions** and precisely mapping the boundary of lazy training in practice; conducting systematic **two-layer width sweeps** to test the $O(1/n) + O(1/r)$ finite-width corrections suggested by the expansion and identify minimal (n, r) configurations where kernel predictions become accurate; and a detailed **comparison suite with MLPs** that probes NTK eigenvalue distributions, validates the spike-bulk structure, and measures task accuracy at matched parameter counts. These studies would verify the **exponential concentration** rates (Theorem 4.1), validate the **single-outlier structure** (Theorem 5.2), and demonstrate the **practical accessibility of the kernel regime** for low-rank random features.

Looking further ahead, we are hopeful that future versions of this work will build toward a **complete unified theory** analogous to tensor programs [50] but specifically tailored for low-rank random feature architectures. Such a framework would provide systematic computational rules for NTK analysis across **different low-rank regimes** (varying r and scaling relationships), arbitrary **layer depths**, and heterogeneous width configurations. This would ultimately yield a comprehensive theoretical foundation for understanding finite-width effects, feature learning, and optimization dynamics in low-rank random-feature networks, with potential applications to architecture design, hyperparameter tuning, and understanding generalization.

Our present scope is deliberately focused on the **lazy/NTK regime**, where analytical tractability is greatest and the advantages of low-rank structure are most clearly visible. We believe that completing the pending finite-width formulas, spectrum characterizations, and experimental validations will turn this into a comprehensive NTK account of RF-LR networks and establish a solid foundation for analyses beyond the lazy regime.

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8 Proofs of NTK Recursions

8.1 Proof of Theorem 3.1: Infinite-width NTK recursion

Proof. We prove the recursion by decomposing the NTK at layer ℓ into contributions of the trainable parameters up to layer ℓ, and invoking the Law of Large Numbers (LLN) under the sequential infinite-width limit. Let $f^{(\ell)}$ denote the network output truncated at layer ℓ. For finite width, the (random) NTK reads

$$\Theta^{(\ell)}_{\mathrm{rand}}(x,x') \;=\; \underbrace{\big(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x)\big)^{\top}\big(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x')\big)}_{\text{Term 1}} \;+\; \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A^{(\ell)}_j}\frac{\partial f^{(\ell)}(x')}{\partial A^{(\ell)}_j}}_{\text{Term 2}}.$$

Bias parameters. Under the NTK parameterization with trainable biases at each layer, the gradient with respect to the bias contributes an identity kernel. Concretely, for layer ℓ, the bias partial derivative satisfies $\partial f^{(\ell)}(x)/\partial b^{(\ell)} = \sigma_A/\sqrt{n_\ell}$, so the associated NTK contribution converges by LLN to a deterministic constant, which we normalize as 1 under our parameterization. This yields the additive $+1$ term in the recursion.

Term 2 (layer-ℓ output weights $A^{(\ell)}$). By the forward pass, $\partial f^{(\ell)}(x)/\partial A^{(\ell)}_j = \frac{\sigma_A}{\sqrt{n_\ell}}\,\boldsymbol{\sigma}\big(\boldsymbol{w}^{(\ell)\top}_j \boldsymbol{h}^{(\ell-1)}(x) + \beta b^{(\ell)}_j\big)$. Therefore

$$\text{Term 2} \;=\; \frac{\sigma_A^2}{n_\ell}\sum_{j=1}^{n_\ell} \boldsymbol{\sigma}\big(\boldsymbol{w}^{(\ell)\top}_j \boldsymbol{h}^{(\ell-1)}(x) + \beta b^{(\ell)}_j\big)\;\boldsymbol{\sigma}\big(\boldsymbol{w}^{(\ell)\top}_j \boldsymbol{h}^{(\ell-1)}(x') + \beta b^{(\ell)}_j\big).$$

Conditionally on $\boldsymbol{h}^{(\ell-1)}$, the summands are i.i.d. over $(\boldsymbol{w}^{(\ell)}_j, b^{(\ell)}_j)$. As $n_\ell \to \infty$ (with lower layers already taken to their limits), LLN yields

$$\text{Term 2} \xrightarrow{p} \sigma_A^2\,\Sigma^{(\ell)}(x,x').$$

Term 1 (previous layers). By the chain rule,

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) \;=\; \Big(\frac{\partial f^{(\ell)}(x)}{\partial \boldsymbol{h}^{(\ell-1)}(x)}\Big)\nabla_{\theta^{(\ell-1)}}\boldsymbol{h}^{(\ell-1)}(x).$$

Hence Term 1 factorizes as $\Theta^{(\ell-1)}_{\mathrm{rand}}(x,x')$ multiplied by a Jacobian metric:

$$\text{Term 1} \;=\; \Theta^{(\ell-1)}_{\mathrm{rand}}(x,x') \;\cdot\; \Big\langle \frac{\partial f^{(\ell)}}{\partial \boldsymbol{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \boldsymbol{h}^{(\ell-1)}}(x')\Big\rangle.$$

Taking the inner product and applying LLN as $n_\ell \to \infty$ yields

$$\Big\langle \frac{\partial f^{(\ell)}}{\partial \boldsymbol{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \boldsymbol{h}^{(\ell-1)}}(x')\Big\rangle \xrightarrow{p} \sigma_A^2\,\dot{\Sigma}^{(\ell)}(x,x').$$

Therefore,

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x,x')\cdot\sigma_A^2\,\dot{\Sigma}^{(\ell)}(x,x').$$

Combining the bias contribution with the two limits completes the recursion

$$\Theta^{(\ell)}(x,x') \;=\; 1 \;+\; \Theta^{(\ell-1)}(x,x')\cdot\sigma_A^2\,\dot{\Sigma}^{(\ell)}(x,x') \;+\; \sigma_A^2\,\Sigma^{(\ell)}(x,x').$$

The sequential limit assumption ensures that lower-layer random objects have already converged when taking the next-layer limit, justifying the conditional LLN applications. □

8.2 Proof of Theorem 2.1: Gaussian process composition

Proof. We prove by induction on ℓ that $\boldsymbol{h}^{(\ell)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell)}$.

Base case $\ell = 0$. For $\ell = 0$, we have $\boldsymbol{h}^{(0)}(x) = x$, which is deterministic and trivially a Gaussian process.

Inductive step. Assume that $\boldsymbol{h}^{(\ell-1)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell-1)}$. Consider the forward pass at layer ℓ:

$$\boldsymbol{h}^{(\ell)}(x) = \frac{1}{\sqrt{\boldsymbol{n}_\ell}} \sum_{j=1}^{\boldsymbol{n}_\ell} A_j^{(\ell)} \ \boldsymbol{\sigma}\left(w_j^{(\ell)\top} \boldsymbol{h}^{(\ell-1)}(x) + b_j^{(\ell)}\right) + c.$$

For any finite collection of inputs $\{x_1, \ldots, x_m\}$, conditionally on $\boldsymbol{h}^{(\ell-1)}$, the terms $A_j^{(\ell)} \ \boldsymbol{\sigma}\left(w_j^{(\ell)\top} \boldsymbol{h}^{(\ell-1)}(x_i) + b_j^{(\ell)}\right)$ are independent across j and identically distributed. Since $A_j^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2)$ and $w_j^{(\ell)}, b_j^{(\ell)}$ are frozen Gaussian draws, the conditional distribution of each term is centered with finite variance.

By the multivariate Central Limit Theorem, as $\boldsymbol{n}_\ell \to \infty$, the vector $(\boldsymbol{h}^{(\ell)}(x_1), \ldots, \boldsymbol{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian. The covariance is given by

$$\mathbb{E}\Big[\boldsymbol{h}^{(\ell)}(x_i)\, \boldsymbol{h}^{(\ell)}(x_j)\Big] = \mathbb{E}_w\Big[\ \boldsymbol{\sigma}\left(w^\top \boldsymbol{h}^{(\ell-1)}(x_i)\right)\ \boldsymbol{\sigma}\left(w^\top \boldsymbol{h}^{(\ell-1)}(x_j)\right)\Big] = \Sigma^{(\ell)}(x_i, x_j),$$

where the expectation is taken over the frozen weights w and the limiting Gaussian process $\boldsymbol{h}^{(\ell-1)}$. The last equality follows from the definition of the base kernel in Definition 2.1.

The sequential infinite-width limit ensures that $\boldsymbol{h}^{(\ell-1)}$ has already converged to its Gaussian process limit when taking the limit for layer ℓ, justifying the application of the CLT conditionally on $\boldsymbol{h}^{(\ell-1)}$. □

8.3 Proof of Two-layer NTK Corollary

Proof. Apply Theorem 3.1 for $\ell = 1$ and $\ell = 2$. For $\ell = 1$, since $\boldsymbol{\Theta}^{(0)} = 0$, the recursion gives:

$$\boldsymbol{\Theta}^{(1)}(x, x') = 1 + \boldsymbol{\Theta}^{(0)}(x, x') \cdot \dot{\Sigma}^{(1)}(x, x') + \Sigma^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x').$$

For $\ell = 2$, substitute $\boldsymbol{\Theta}^{(1)} = 1 + \Sigma^{(1)}$ into the recursion:

$$\boldsymbol{\Theta}^{(2)}(x, x') = 1 + \boldsymbol{\Theta}^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \Sigma^{(2)}(x, x').$$

This is the stated two-layer recursion with the bias contribution.

Mean explicit form. Under the homogeneous ReLU setting on the sphere, the last-layer expectations are

$$\mathbb{E}\big[\dot{\Sigma}^{(2)}(\rho)\big] = 1 - \frac{\arccos(\rho)}{\pi}, \qquad \mathbb{E}\big[\Sigma^{(2)}(\rho)\big] = \Sigma^{(1)}(\rho).$$

Taking the conditional mean at layer 2 yields the mean two-layer NTK as a function of ρ:

$$\tilde{\boldsymbol{\Theta}}^{(2)}(\rho) \ = \ 1 \ + \ \boldsymbol{\Theta}^{(1)}(\rho)\left(1 - \frac{\arccos(\rho)}{\pi}\right) \ + \ \Sigma^{(1)}(\rho).$$

Since $\boldsymbol{\Theta}^{(1)}(\rho) = 1 + \Sigma^{(1)}(\rho)$, this can be written as

$$\tilde{\boldsymbol{\Theta}}^{(2)}(\rho) \ = \ \underbrace{\boldsymbol{\Theta}^{(1)}(\rho)}_{\Theta^{(1)}(\rho)}\left(1 - \frac{\arccos(\rho)}{\pi}\right) \ + \ \Sigma^{(1)}(\rho) \ + \ 1 \ = \ K_\infty(\rho),$$

which proves the explicit expression. □

8.4 Proof of General L-layer NTK Corollary

Proof. We prove the explicit form by induction on L, accounting for the +1 bias terms in the recursion.

Base case $L = 1$. From Theorem 3.1, $\boldsymbol{\Theta}^{(1)} = 1 + \Sigma^{(1)}$. The formula with $L = 1$ gives:

$$\boldsymbol{\Theta}^{(1)}(x, x') = \sum_{\ell=1}^{1} \left(\prod_{k=\ell+1}^{1} \dot{\Sigma}^{(k)}(x, x')\right) + \sum_{\ell=1}^{1} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{1} \dot{\Sigma}^{(k)}(x, x').$$

For the first sum: when $\ell = 1$, $\prod_{k=2}^{1} \dot{\Sigma}^{(k)} = 1$ (empty product). For the second sum: when $\ell = 1$, $\Sigma^{(1)} \cdot \prod_{k=2}^{1} \dot{\Sigma}^{(k)} = \Sigma^{(1)} \cdot 1 = \Sigma^{(1)}$. Therefore, $\boldsymbol{\Theta}^{(1)} = 1 + \Sigma^{(1)}$, which matches the recursion.

Inductive step. Assume the formula holds for depth $L-1$:

$$\mathbf{\Theta}^{(L-1)}(x,x') = \sum_{\ell=1}^{L-1}\left(\prod_{k=\ell+1}^{L-1}\dot{\Sigma}^{(k)}(x,x')\right) + \sum_{\ell=1}^{L-1}\Sigma^{(\ell)}(x,x')\prod_{k=\ell+1}^{L-1}\dot{\Sigma}^{(k)}(x,x').$$

By Theorem 3.1,

$$\mathbf{\Theta}^{(L)}(x,x') = 1 + \mathbf{\Theta}^{(L-1)}(x,x')\cdot\dot{\Sigma}^{(L)}(x,x') + \Sigma^{(L)}(x,x').$$

Substituting the inductive hypothesis:

$$\begin{aligned}
\mathbf{\Theta}^{(L)}(x,x') &= 1 + \left[\sum_{\ell=1}^{L-1}\left(\prod_{k=\ell+1}^{L-1}\dot{\Sigma}^{(k)}(x,x')\right) + \sum_{\ell=1}^{L-1}\Sigma^{(\ell)}(x,x')\prod_{k=\ell+1}^{L-1}\dot{\Sigma}^{(k)}(x,x')\right]\cdot\dot{\Sigma}^{(L)}(x,x') + \Sigma^{(L)}(x,x') \\
&= 1 + \sum_{\ell=1}^{L-1}\left(\prod_{k=\ell+1}^{L}\dot{\Sigma}^{(k)}(x,x')\right) + \sum_{\ell=1}^{L-1}\Sigma^{(\ell)}(x,x')\prod_{k=\ell+1}^{L}\dot{\Sigma}^{(k)}(x,x') + \Sigma^{(L)}(x,x').
\end{aligned}$$

The first term 1 corresponds to $\ell = L$ in the first sum (with empty product $\prod_{k=L+1}^{L}\dot{\Sigma}^{(k)} = 1$). The last term $\Sigma^{(L)}$ corresponds to $\ell = L$ in the second sum (with empty product $\prod_{k=L+1}^{L}\dot{\Sigma}^{(k)} = 1$). Therefore:

$$\mathbf{\Theta}^{(L)}(x,x') = \sum_{\ell=1}^{L}\left(\prod_{k=\ell+1}^{L}\dot{\Sigma}^{(k)}(x,x')\right) + \sum_{\ell=1}^{L}\Sigma^{(\ell)}(x,x')\prod_{k=\ell+1}^{L}\dot{\Sigma}^{(k)}(x,x'),$$

completing the induction. The total number of terms is $L + L = 2L$, which grows linearly with depth. However, when considering finite-width corrections and interactions between terms, the complexity becomes quadratic. □

9 Proofs of low-rank NTK RKHS

9.1 Proof of Theorem 4.1: Rank-driven concentration

Proof. Let $x_1, y_1 \in \mathbb{R}^r$ be Gaussian projections with arbitrary fixed correlation $\rho \in [-1,1]$. Define $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$, and $W = UV = \|x_1\|\,\|y_1\|/r$. We prove that for $\epsilon \in (0,1)$,

$$\mathbb{P}(|W-1| \geq \epsilon) \leq 4\exp\left(-\frac{r\epsilon^2}{8}\right),$$

and that for $\epsilon \geq 1$ there exist constants $c_1, c_2 > 0$ with $\mathbb{P}(|W-1| \geq \epsilon) \leq c_1 e^{-c_2 r\epsilon}$.

The Euclidean norm is 1-Lipschitz, so by Gaussian concentration, for all $\delta > 0$,

$$\mathbb{P}(|U-1| \geq \delta) \leq 2\exp\left(-\frac{r\delta^2}{2}\right), \qquad \mathbb{P}(|V-1| \geq \delta) \leq 2\exp\left(-\frac{r\delta^2}{2}\right).$$

We use

$$|UV-1| \;\leq\; |U-1| + |V-1| + |U-1|\,|V-1|.$$

Fix $\epsilon \in (0,1)$ and set $\delta = \epsilon/2$. On the event $\{|U-1| < \delta,\ |V-1| < \delta\}$, we have

$$|UV-1| \;\leq\; 2\delta + \delta^2 \;\leq\; \epsilon.$$

Therefore,

$$\mathbb{P}(|UV-1| \geq \epsilon) \;\leq\; \mathbb{P}(|U-1| \geq \delta) \;+\; \mathbb{P}(|V-1| \geq \delta) \;\leq\; 4\exp\left(-\frac{r\epsilon^2}{8}\right).$$

□

9.2 Proof of Corollary 4.1: Concentration bound for two-layer NTK

Proof. We prove that the empirical two-layer NTK $\hat{\Theta}_r^{(2)}(x,x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ concentrates exponentially fast around its deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$ as $r \to \infty$.

Setup. Recall that the kernel function is defined as:

$$\Psi(u,w) = \Theta^{(1)}(u)\left(1 - \frac{\arccos(u)}{\pi}\right) + w\,\Sigma^{(1)}(u) + 1,$$

where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\|\|y_1\|/r$. By Fisher's CLT [7, 8], $\sqrt{r}(\hat{\rho}_r - \rho) \xrightarrow{d} \mathcal{N}(0,(1-\rho^2)^2)$, and by Theorem 4.1, $\hat{w}_r$ concentrates around 1 with sub-Gaussian tails.

Lipschitz properties of Ψ. The function Ψ is differentiable in both arguments. For fixed $\rho \in (-1,1)$, we have:

$$\partial_u \Psi(\rho,1) = \partial_\rho K_\infty(\rho) = O(1), \tag{8}$$

$$\partial_w \Psi(\rho,1) = \Sigma^{(1)}(\rho) = O(1). \tag{9}$$

Since $\Theta^{(1)}$, $\Sigma^{(1)}$, and arccos are smooth on $(-1,1)$, there exist Lipschitz constants $L_u, L_w > 0$ (depending on ρ) such that for $u,u' \in [\rho-\delta,\rho+\delta]$ and $w,w' \in [1-\delta,1+\delta]$ with $\delta > 0$ small:

$$|\Psi(u,w) - \Psi(u',w')| \le L_u|u-u'| + L_w|w-w'|.$$

Combining radial and angular concentration. By a union bound and the Lipschitz property:

$$|\hat{\Theta}_r^{(2)}(x,x') - K_\infty(\rho)| = |\Psi(\hat{\rho}_r,\hat{w}_r) - \Psi(\rho,1)| \le L_u|\hat{\rho}_r - \rho| + L_w|\hat{w}_r - 1|.$$

Therefore, for $\epsilon \in (0,1)$:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x,x') - K_\infty(\rho)| \ge \epsilon\right) \le \mathbb{P}\left(|\hat{\rho}_r - \rho| \ge \frac{\epsilon}{2L_u}\right) + \mathbb{P}\left(|\hat{w}_r - 1| \ge \frac{\epsilon}{2L_w}\right).$$

Angular concentration (Fisher). By Fisher's CLT [7, 8], $\sqrt{r}(\hat{\rho}_r - \rho) \xrightarrow{d} \mathcal{N}(0,(1-\rho^2)^2)$. For large r, using sub-Gaussian concentration:

$$\mathbb{P}\left(|\hat{\rho}_r - \rho| \ge \frac{\epsilon}{2L_u}\right) \le 2\exp\left(-\frac{r\epsilon^2}{8L_u^2(1-\rho^2)^2}\right).$$

Radial concentration (Theorem 4.1). By Theorem 4.1, for $\epsilon/(2L_w) \in (0,1)$:

$$\mathbb{P}\left(|\hat{w}_r - 1| \ge \frac{\epsilon}{2L_w}\right) \le 4\exp\left(-\frac{r\epsilon^2}{32L_w^2}\right).$$

Final bound. Combining both terms:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x,x') - K_\infty(\rho)| \ge \epsilon\right) \le 2\exp\left(-\frac{r\epsilon^2}{8L_u^2(1-\rho^2)^2}\right) + 4\exp\left(-\frac{r\epsilon^2}{32L_w^2}\right).$$

Taking $C_1 = 6$ and $C_2 = \min(1/(8L_u^2(1-\rho^2)^2), 1/(32L_w^2))$, we obtain the stated exponential concentration bound. □

9.3 Fisher and Kibble Distributions

Lemma 9.1 (Fisher–Kibble decoupling)**.** *Let x, y be input vectors and let x_1, y_1 denote their rank-r random projections. Define the empirical correlation $\rho_1 = \cos\angle(x_1,y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- *ρ_1 follows Fisher's correlation distribution [7] [8], centered at the true correlation ρ.*
- *(u,v) follow Kibble's [44] bivariate Gamma (chi-square) law.*
- *Angular and radial parts are independent: $p(\rho_1,u,v) = p_{Fisher}(\rho_1)\,p_{Kibble}(u,v)$.*

Proof. Let $x, y \in \mathbb{R}^d$ be fixed input vectors with correlation $\rho = \langle x,y\rangle/(\|x\|\|y\|)$. Consider a rank-$r$ random projection matrix $P \in \mathbb{R}^{r\times d}$ with entries $P_{ij} \sim \mathcal{N}(0,1/d)$ i.i.d., so that $x_1 = Px$ and $y_1 = Py$ are the projected vectors.

Since P has i.i.d. Gaussian entries, the projected vectors $x_1, y_1 \in \mathbb{R}^r$ form a ρ-correlated bivariate Gaussian pair. Specifically, for each component $i = 1,\ldots,r$, the pairs (x_{1i}, y_{1i}) are i.i.d. with joint distribution

$$\begin{pmatrix} x_{1i} \\ y_{1i} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \|x\|^2/d & \rho\|x\|\|y\|/d \\ \rho\|x\|\|y\|/d & \|y\|^2/d \end{pmatrix}\right).$$

By rotational invariance of the Gaussian projection and the fact that the correlation ρ is preserved under orthogonal transformations, we can assume without loss of generality that the covariance structure factors into angular and radial components.

Angular part (Fisher distribution). The empirical correlation is defined as

$$\rho_1 = \frac{\langle x_1, y_1 \rangle}{\|x_1\| \|y_1\|} = \frac{\sum_{i=1}^r x_{1i} y_{1i}}{\sqrt{\sum_{i=1}^r x_{1i}^2} \sqrt{\sum_{i=1}^r y_{1i}^2}}.$$

This is the sample correlation coefficient of r pairs of correlated Gaussian random variables. Fisher [7, 8] showed that the distribution of ρ_1 depends only on the true correlation ρ and the sample size r, with density given in Theorem 3.1. The key observation is that ρ_1 is a function purely of the angles between the projected vectors, independent of their magnitudes.

Radial part (Kibble distribution). The squared norms $u = \|x_1\|^2 = \sum_{i=1}^r x_{1i}^2$ and $v = \|y_1\|^2 = \sum_{i=1}^r y_{1i}^2$ are sums of correlated chi-square variables. Since (x_{1i}, y_{1i}) are ρ-correlated Gaussians, the bivariate distribution of (u, v) follows Kibble's [44] bivariate chi-square law. The joint density involves the modified Bessel function $I_\nu(z)$, which appears because the dot product $\langle x_1, y_1 \rangle = \sum_{i=1}^r x_{1i} y_{1i}$ can be expressed as a weighted sum of products of correlated normals, whose distribution relates to Bessel functions through the generating function of the bivariate chi-square distribution.

Specifically, the joint characteristic function of (u, v) is

$$\phi(s,t) = \mathbb{E}[e^{i(su+tv)}] = \left(1 - 2i(1-\rho^2)(s+t) - 4\rho^2 st\right)^{-r/2},$$

and the inverse Fourier transform yields Kibble's density, which contains the modified Bessel function $I_{\frac{r}{2}-1}$ as shown in Lemma 3.1. The Bessel function arises from the integral representation of the bivariate chi-square density when the correlation $\rho \neq 0$.

Independence of angular and radial parts. Rotational invariance of isotropic Gaussian projections implies that $\rho_1 = \cos\angle(x_1, y_1)$ and the norms $(\|x_1\|, \|y_1\|)$ are independent, yielding the factorization $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) \cdot p_{\text{Kibble}}(u, v)$. This is the classical independence of the sample correlation from sample variances in bivariate normal data. □

9.4 Proof of Corollary 4.2: Mean NTK over Fisher–Kibble has same RKHS

Proof. We prove that under Assumption 2.1, the mean NTK $\tilde{\mathbf{\Theta}}^{(L)}$ (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow (two-layer) ReLU kernel.

Puiseux expansion of mean NTK near $\rho = 1$: Full Fisher density computation. The key observation is that the mean NTK $\tilde{\mathbf{\Theta}}^{(2)}(\rho)$ is a zonal kernel (depends only on ρ). Near $\rho = 1$ (for $t \geq 0$), we need to compute the Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\rho = 1 - t$, where $\hat{\rho}_r \sim \text{Fisher}(1-t, r)$.

This is highly non-trivial because we must compute:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_{-1}^{1} \arccos(u)\, p_{\text{Fisher}}(u \mid \rho = 1-t, r)\, du,$$

where the full Fisher distribution density is:

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2)\,\Gamma(r-1)\,(1-\rho^2)^{\frac{r-1}{2}}\,(1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(1-\rho u)^{r-\frac{3}{2}}}\, {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2};\, r - \tfrac{1}{2};\, \tfrac{1+\rho u}{2}\right),$$

for $r > 2$, with ${}_2F_1$ the hypergeometric function.

Making the change of variables $v = 1 - u$ and $s = 1 - t$, so $\rho = s = 1 - t$ and $u = 1 - v$, we have:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^2 \arccos(1-v)\, p_{\text{Fisher}}(1-v \mid s, r)\, dv.$$

Near $v = 0$ and $t = 0$, we analyze the asymptotic behavior of the Fisher density. For $\rho = s = 1 - t$ and $u = 1 - v$ with $t, v \to 0^+$:

$$1 - \rho^2 = 1 - (1-t)^2 = 2t - t^2 = 2t(1 - t/2), \tag{10}$$

$$1 - u^2 = 1 - (1-v)^2 = 2v - v^2 = 2v(1 - v/2), \tag{11}$$

$$1 - \rho u = 1 - (1-t)(1-v) = t + v - tv = t + v + O(tv). \tag{12}$$

The Fisher density near $v = 0$ and $t = 0$ behaves as:

$$p_{\text{Fisher}}(1 - v \mid 1 - t, r) \sim \frac{(r-2)\,\Gamma(r-1)\,(2t)^{\frac{r-1}{2}}\,(2v)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(t+v)^{r-\frac{3}{2}}}\,{}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2};\, r - \tfrac{1}{2};\, 1 - \tfrac{t+v}{2}\right).$$

For the hypergeometric function near its argument $1 - (t+v)/2 \to 1^-$, we use the connection formula and asymptotic expansion from [47] and [48]:

$${}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2};\, r - \tfrac{1}{2};\, 1 - \tfrac{t+v}{2}\right) \sim C_1(r) + C_2(r)(t+v) + O((t+v)^2),$$

where $C_1(r), C_2(r)$ are constants depending on r but not on t or v.

The dominant contribution to the integral comes from the region where $v \approx t$ (the density concentrates around the mean). Near this region, $\arccos(1 - v) = \sqrt{2v} + O(v^{3/2})$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2v} \cdot \frac{(r-2)\,\Gamma(r-1)\,(2t)^{\frac{r-1}{2}}\,(2v)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(t+v)^{r-\frac{3}{2}}}\,C_1(r)\,dv + O(t^{3/2}).$$

Making the substitution $w = v/t$, we have $v = tw$, $dv = t\,dw$, and $t + v = t(1+w)$:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2tw} \cdot \frac{(r-2)\,\Gamma(r-1)\,(2t)^{\frac{r-1}{2}}\,(2tw)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(t(1+w))^{r-\frac{3}{2}}}\,C_1(r) \cdot t\,dw + O(t^{3/2}).$$

Simplifying the powers of t:

$$= \sqrt{2t} \cdot t^{\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r - \frac{3}{2})} \cdot \frac{(r-2)\,\Gamma(r-1)\,(2)^{\frac{r-1}{2}}\,(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(1+w)^{r-\frac{3}{2}}}\,C_1(r) \cdot \sqrt{w}\,dw + O(t^{3/2}),$$

where the exponent is $\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r - \frac{3}{2}) = 0$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} \cdot I(r) + O(t^{3/2}),$$

where

$$I(r) = \int_0^\infty \sqrt{w} \cdot \frac{(r-2)\,\Gamma(r-1)\,(2)^{\frac{r-1}{2}}\,(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\,\Gamma(r-\frac{1}{2})\,(1+w)^{r-\frac{3}{2}}}\,C_1(r)\,dw.$$

Evaluating the integral yields the exact constant

$$I(r) \;=\; \frac{(r-2)\,2^{r-\frac{5}{2}}\,\Gamma\!\left(\frac{r-1}{2}\right)\Gamma\!\left(\frac{r}{2} - 1\right)}{\sqrt{2\pi}\,\Gamma(r-1)}\,C_1(r), \qquad C_1(r) \;=\; \frac{\Gamma\!\left(r - \frac{1}{2}\right)\Gamma\!\left(r - \frac{3}{2}\right)}{\Gamma(r-1)^2}.$$

Therefore,

$$\mathbb{E}[\arccos(\hat{\rho}_r)] \;=\; \sqrt{2t}\,I(r) \;+\; O(t^{3/2}).$$

Therefore, the mean NTK expansion is:

$$\tilde{\boldsymbol{\Theta}}^{(2)}(1-t) \;=\; K_\infty(1) \;-\; \left[\frac{2}{\pi}\,\frac{\sqrt{2}}{\pi}\,\Theta^{(1)}(1)\,\frac{I(r)}{2}\right] t^{1/2} \;+\; O(t^{3/2}),$$

so, for ReLU with $\Theta^{(1)}(1) = \frac{1}{2}$,

$$\tilde{\boldsymbol{\Theta}}^{(2)}(1-t) \;=\; K_\infty(1) \;-\; \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi}\,I(r)\right] t^{1/2} \;+\; O(t^{3/2}).$$

This exhibits the complete r-dependent constant multiplying $t^{1/2}$.

RKHS equivalence via endpoint behavior. The mean NTK $\tilde{\mathbf{\Theta}}^{(2)}(\rho)$ is a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$ due to the Fisher–Kibble expectations. However, the leading-order behavior near $\rho = 1$ is preserved: both kernels have the same $t^{1/2}$ leading term in their Puiseux expansions.

By Theorem 1 of [25], the RKHS spectral decay is determined by the leading-order Puiseux expansion coefficient (the $t^{1/2}$ term). Since $\tilde{\mathbf{\Theta}}^{(2)}(\rho)$ and the shallow ReLU kernel share the same leading-order expansion $1 - \frac{2}{\pi}t^{1/2} + \cdots$ (with different higher-order corrections), they induce the same RKHS. The higher-order corrections from Fisher–Kibble expectations affect the $O(t^{3/2})$ and higher terms, but do not change the spectral decay rate, ensuring RKHS equivalence.

Extension to general L. For general depth L, the mean NTK involves nested expectations over Fisher and Kibble distributions at each layer. The key is that homogeneity ensures the endpoint Puiseux expansions are preserved (up to $O(r^{-1/2})$ corrections per layer), so the RKHS structure remains the same. Proofs is deferred to a future version. □

10 Spectral Theory

10.1 Asymptotic Signal-Noise Decomposition

Proof of Theorem 5.1. We establish the asymptotic distribution of the empirical kernel $\hat{\Theta}_r^{(2)} = \Psi(\hat{\rho}_r, \hat{w}_r)$ via a joint CLT and the multivariate Delta method. Let $\mathbf{Z}_r = (\hat{\rho}_r,\ \hat{w}_r)^\top$. For isotropic Gaussian projections in rank r, Fisher's theory [7, 8] gives

$$\sqrt{r}\,(\hat{\rho}_r - \rho) \ \xrightarrow{d}\ \mathcal{N}\Big(0,\ (1-\rho^2)^2\Big).$$

Writing $u = \|x\|^2$, $v = \|y\|^2$ and $\hat{w}_r = \sqrt{uv}/r$, Kibble's bivariate chi-square yields

$$\sqrt{r}\,(\hat{w}_r - 1) \ \xrightarrow{d}\ \mathcal{N}\Big(0,\ 1+\rho^2\Big).$$

Angular and radial components are independent for isotropic Gaussians, hence

$$\sqrt{r}\,\Big(\mathbf{Z}_r - (\rho,1)^\top\Big) \ \xrightarrow{d}\ \mathcal{N}\Big(\mathbf{0},\ \mathbf{\Lambda}(\rho)\Big), \quad \mathbf{\Lambda}(\rho) \ = \ \begin{pmatrix} (1-\rho^2)^2 & 0 \\ 0 & 1+\rho^2 \end{pmatrix}.$$

The map $\Psi(u,w) = \Theta^{(1)}(u)\big(1 - \arccos(u)/\pi\big) + w\,\Sigma^{(1)}(u) + 1$ is C^1 near $(\rho,1)$. Therefore,

$$\sqrt{r}\,\Big(\Psi(\mathbf{Z}_r) - \Psi(\rho,1)\Big) \ \xrightarrow{d}\ \mathcal{N}\Big(0,\ \nabla\Psi(\rho,1)^\top\mathbf{\Lambda}(\rho)\,\nabla\Psi(\rho,1)\Big).$$

We have $\partial_u\Psi(\rho,1) = \partial_\rho K_\infty(\rho)$ and $\partial_w\Psi(\rho,1) = \Sigma^{(1)}(\rho)$. Using the diagonal covariance, we obtain

$$\sigma^2_{\mathrm{tot}}(\rho) \ = \ \big(\partial_\rho K_\infty(\rho)\big)^2\,(1-\rho^2)^2 \ + \ \big(\Sigma^{(1)}(\rho)\big)^2\,(1+\rho^2).$$

Hence

$$\hat{\Theta}_r^{(2)} \ = \ K_\infty(\rho) \ + \ \frac{1}{\sqrt{r}}\,\mathcal{G}(\rho) \ + \ o_P(r^{-1/2}), \quad \mathcal{G}(\rho) \sim \mathcal{N}\big(0, \sigma^2_{\mathrm{tot}}(\rho)\big),$$

which proves the claim. □

10.2 Proof of Theorem 5.2: Deformed Marchenko–Pastur with Spike

Proof. We prove the spectral limit using Theorem 2.1 of [49] on the spectrum of inner product kernel random matrices. This connects the microscopic kernel structure (ReLU/Kibble calculations) with the macroscopic matrix behavior (spectrum).

Setting. Let $\mathbf{M} = [\Theta_{ij}^{(2)}]_{i,j=1}^n$ be the $n \times n$ NTK matrix with entries $M_{ij} = \Theta^{(2)}(x_i, x_j) = K_\infty(\rho_{ij}) + O_P(r^{-1/2})$, where $\rho_{ij} = \langle x_i, x_j\rangle/(\|x_i\|\|x_j\|)$ is the cosine similarity and $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ is the deterministic two-layer kernel limit.

Under Assumption 5.1, we have $n \sim r$ (so n/p remains bounded where p is the ambient dimension) and the data points x_i are normalized. The kernel function $f(\rho) = K_\infty(\rho)$ is C^1 in a neighborhood of $l = \lim_{p\to\infty} \mathrm{trace}(\Sigma_p)/p = 1$ (for normalized data) and C^3 in a neighborhood of 0, satisfying the regularity conditions of [49].

Step 1: Equivalent Matrix Structure. By Theorem 2.1 of [49], in probability as $n, p \to \infty$, the kernel matrix $\mathbf{M}$ is equivalent in operator norm to the matrix $\mathbf{K}$:

$$\mathbf{K} = \underbrace{\alpha \mathbf{1}\mathbf{1}^T}_{\text{Spike}} + \underbrace{\beta \frac{XX^T}{p}}_{\text{Bulk}} + \underbrace{\gamma I_n}_{\text{Shift}},$$

where $X = [x_1, \ldots, x_n] \in \mathbb{R}^{p\times n}$ is the data matrix. The three spectral coefficients α, β, γ are determined by the kernel function $f(\rho) = K_\infty(\rho)$ and its derivatives evaluated at $\rho = 0$ and $\rho = 1$:

$$\alpha = f(0) = K_\infty(0), \tag{13}$$

$$\beta = f'(0) = K'_\infty(0), \tag{14}$$

$$\gamma = f(1) - f(0) - f'(0) = K_\infty(1) - K_\infty(0) - K'_\infty(0). \tag{15}$$

Step 2: Calculation of Spectral Coefficients. A. The Spike (α): The Outlier. The spike coefficient is the kernel value at orthogonality ($\rho = 0$):

$$\alpha = K_\infty(0) = \Theta^{(1)}(0)\left(1 - \frac{\arccos(0)}{\pi}\right) + \Sigma^{(1)}(0) + 1.$$

Since $\arccos(0) = \pi/2$, we obtain:

$$\alpha = \Theta^{(1)}(0)\left(1 - \frac{\pi}{2\pi}\right) + \Sigma^{(1)}(0) + 1 = \Theta^{(1)}(0)\left(1 - \frac{1}{2}\right) + \Sigma^{(1)}(0) + 1 = \frac{\Theta^{(1)}(0)}{2} + \Sigma^{(1)}(0) + 1.$$

For ReLU activations with normalized inputs, $\Theta^{(1)}(0) = \Sigma^{(1)}(0) = 0$, hence $\alpha = 1$. The rank-one mean term $\alpha\mathbf{1}\mathbf{1}^T$ induces an outlier eigenvalue:

$$\lambda_{\text{spike}} = n \cdot \alpha = n \cdot K_\infty(0),$$

of order $O(n)$, isolated from the bulk spectrum.

B. The Bulk (β): The Linear Scale. The bulk coefficient is the derivative of the kernel at $\rho = 0$, capturing the linear sensitivity:

$$\beta = K'_\infty(0) = \frac{d}{d\rho}\left[\Theta^{(1)}(\rho)\left(1 - \frac{\arccos(\rho)}{\pi}\right) + \Sigma^{(1)}(\rho) + 1\right]_{\rho=0}.$$

Differentiating term by term:

$$\frac{d}{d\rho}\left[\Theta^{(1)}(\rho)\left(1 - \frac{\arccos(\rho)}{\pi}\right)\right] = \Theta^{(1)\prime}(\rho)\left(1 - \frac{\arccos(\rho)}{\pi}\right) + \Theta^{(1)}(\rho) \cdot \frac{1}{\pi\sqrt{1-\rho^2}},$$

$$\frac{d}{d\rho}\left[\Sigma^{(1)}(\rho)\right] = \Sigma^{(1)\prime}(\rho).$$

Evaluating at $\rho = 0$ (where $\arccos(0) = \pi/2$ and $\sqrt{1-0} = 1$):

$$\beta = \Theta^{(1)\prime}(0)\left(1 - \frac{1}{2}\right) + \Theta^{(1)}(0) \cdot \frac{1}{\pi} + 2\Sigma^{(1)\prime}(0) = \frac{\Theta^{(1)\prime}(0)}{2} + \frac{\Theta^{(1)}(0)}{\pi} + 2\Sigma^{(1)\prime}(0).$$

For ReLU kernels with normalized inputs, $\Theta^{(1)}(0) = 0$ and $\Theta^{(1)\prime}(0) = 1/\pi$, $\Sigma^{(1)\prime}(0) = 1/(2\pi)$, hence $\beta = 1/(2\pi) + 1/\pi = 3/(2\pi)$.

The term $\beta XX^T/p$ has the same spectral distribution as a Wishart matrix scaled by β. When $n \asymp p$ with aspect ratio $\gamma_{\text{ratio}} = n/r$, the bulk eigenvalues follow a deformed Marchenko–Pastur distribution scaled by β, supported on:

$$\left[\ \beta(1 - \sqrt{\gamma_{\text{ratio}}})^2,\ \ \beta(1 + \sqrt{\gamma_{\text{ratio}}})^2\ \right].$$

C. The Shift (γ): The Diagonal Residual. The shift coefficient accounts for the difference between the diagonal value and the linear approximation:

$$\gamma = K_\infty(1) - K_\infty(0) - K'_\infty(0) = \alpha + \beta \cdot 1 + \text{non-linear terms} - \alpha - \beta = \text{non-linear residual}.$$

Computing $K_\infty(1)$ (on-diagonal, where $\arccos(1) = 0$):

$$K_\infty(1) = \Theta^{(1)}(1) \cdot 1 + \Sigma^{(1)}(1) + 1 = \Theta^{(1)}(1) + \Sigma^{(1)}(1) + 1.$$

Therefore:

$$\gamma = \left[\Theta^{(1)}(1) + \Sigma^{(1)}(1) + 1\right] - \alpha - \beta.$$

For normalized ReLU kernels, $\Theta^{(1)}(1) = \Sigma^{(1)}(1) = 1/2$, so $K_\infty(1) = 1 + 1 + 1 = 3$, and $\gamma = 3 - 1 - 3/(2\pi) = 2 - 3/(2\pi) > 0$.

The diagonal shift γI_n adds γ to all eigenvalues, acting as natural Tikhonov regularization. This ensures the kernel matrix is well-conditioned (bounded away from zero) and stabilizes inversion.

Step 3: Final Spectral Distribution. Combining the three components, the spectrum of $\mathbf{M}$ converges (in probability) to:

- **Spike (rank-one outlier):** A single eigenvalue $\lambda_{\text{spike}} = n \cdot \alpha = n \cdot K_\infty(0)$ of order $O(n)$, isolated from the bulk.
- **Bulk (deformed Marchenko–Pastur):** The remaining $n - 1$ eigenvalues follow a deformed MP distribution, supported on:
$$\left[\ \gamma + \beta(1 - \sqrt{\gamma_{\text{ratio}}})^2,\ \ \gamma + \beta(1 + \sqrt{\gamma_{\text{ratio}}})^2\ \right],$$
where $\gamma_{\text{ratio}} = \lim_{n\to\infty} n/r \in (0, \infty)$. The bulk width is determined by $\beta = K'_\infty(0)$, and the bulk location is shifted by $\gamma = K_\infty(1) - K_\infty(0) - K'_\infty(0)$.

Interpretation: Fisher–Kibble Decoupling in the Spectrum. This spectral decomposition reveals how the microscopic Fisher–Kibble structure manifests macroscopically:

- **Fisher's angular fluctuations** (controlled by ρ) determine the bulk scale β through $K'_\infty(0)$, governing learning capacity and conditioning.
- **Kibble's radial fluctuations** (norm variances) concentrate into the deterministic shift γ through the diagonal term $K_\infty(1)$, stabilizing the matrix inversion.
- The **bias term** in the kernel (the +1) propagates directly into α, guaranteeing a dominant spike regardless of data structure.

The theorem is proved. $\square$